



A novel Deep Learning Framework (DLF) for classification of Acute Lymphoblastic Leukemia

Sunita Chand^{1,2} · Virendra P. Vishwakarma³

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Abstract

Cancer of blood, more specifically known as Leukemia, is a deadly disease that is responsible for the abnormal proliferation of immature white blood cells in bone marrow. Due to sudden surge in the speed of disease progression and lack of timely diagnosis, the chance of disease remission also diminish. Leukemia incidence rate has seen a steep rise of 144.7% in past 22 years from 28,700 in 1998 to 60,530 in 2020 (Hao et al. Sci Rep 9(1):1–13, 2019). Deep learning has revolutionised the solution to classification problems, more specifically, image based classification. Thus, in this paper, we propose a novel deep learning framework(DLF) based on convolution neural network for the diagnosis of Acute Lymphoblastic Leukemia (ALL), which is one of the four types of leukemia. The proposed method does not require feature extraction. Also it does not require any pre-training on any other database and thus can be used for real time application for detection of leukemia. The proposed framework is simple compared to existing deep networks. Number of free parameters to be tuned in the proposed framework is 41,626, which is quite less than the learnable parameters in the existing pre-trained complex networks such as, AlexNet (over 60 millions), Visual Geometry Group(VGG-Net)(138 million), Residual Network(ResNet 152)(60.3 millions). As the number of free parameters of the proposed framework is approximately 1400 times less than those of existing deep networks, the simulation of the proposed framework has been possible on simple processor(i5 @2.53GHz processor with 4 GB RAM), it does not require any GPU for processing. Despite of lesser number of free parameters in the proposed model, it is able to diagnose the disease with 100% accuracy in most of the repetitions and an average accuracy of 98.17% .This is the average of two other averages i.e., Experiment-A (98.62%), obtained when the data has been partitioned on (80%-20%) training and testing sets for 20 epochs and Experiment-B(97.73%) when the data has been partitioned as 60-48 training and testing images for 30 epochs.

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✉ Sunita Chand
sunita.usict.131164@ipu.ac.in

¹ USIC&T, Guru Gobind Singh Indraprastha University, New Delhi, India

² Hansraj College, University of Delhi, Delhi, India

³ USICT, Guru Gobind Singh Indraprastha University, New Delhi, India

Keywords Acute lymphoblastic leukemia · Deep learning · Convolutional neural network · Training of CNN · Image based classification

1 Introduction

The fatal nature of cancer is not hidden from anyone. As per estimation from American Cancer Society [9], approximately 19 lac new cases of cancer have been predicted in 2021 in America. The report also predict 608,570 Americans to die of cancer in 2021, which is equivalent to 1670 deaths per day. For leukemia, 61,090 new cases are expected for the current year while expected deaths are reported to be 23,660. According to a cancer statistics published by National Cancer Registry Program, India, for the year 2020, 1 out of every 9 Indians will suffer from cancer malignancies in their lifespan i.e. from birth till 74 years of age [28]. Unlike other cancers, which originate and affect tissues and organs of human body, Leukemia affects blood and hence, is also known as blood cancer. In case of metastasis, the cancer reaches other organs as well, a stage, when it becomes life threatening. Human blood comprises of different types of cells, namely,

- 1) Red Blood Cell (RBCs) or erythrocytes, which helps in gas exchange and provide energy to our body cells.
- 2) White Blood Cells(WBCs) or Leukocytes, which helps in fighting with infections and thus develop immunity, and
- 3) Platelets or Thrombocytes(THR), that helps coagulation of blood in case of any cut or injuries [12].

All the three types of cells originate at the core of our bones in a spongy tissue known as bone-marrow, as stem cells. These stem cells, being pluripotent in nature, have the capability to differentiate into any cell type. At level one, these cells differentiates on the basis of type of cell lineage into myeloblasts or lymphoblasts which act as precursors of blood cells belonging to the respective lineages. For example erythrocytes, thrombocytes and granulocytes (neutrophils, eosinophils and basophils) originate in myeloid cell lineage whereas leukocyte (B-cell, T-Cell and Natural Killer Cells) emerges from lymphoid cell lineage. B-Lymphocytes, also known as B-Cell are called so as they mature in bone marrow, whereas T-Lymphocytes, also known as T-Cells mature in thymus hence named so. Out of these, granulocytes and lymphocytes form are the WBCs. The Fig. 1 depicts the blood cell differentiation representing all types of blood components.

Hence Leukemia may be classified into myeloid leukemia or lymphoid leukemia on the basis of origin of the blasts cells. Not only this, Leukemia may also be classified into acute or chronic on the basis of the speed of its progression [8, 10]. Thus the term ‘Leukemia’ represents a collection of various types of cancers i.e., Acute myeloid Leukemia(AML), Acute Lymphoid Leukemia(ALL) , Chronic myeloid Leukemia(CML), Chronic Lymphoid Leukemia(CLL) and hence is heterogeneous in nature and demands different treatments for different types [51]. Leukemia is suspected in the preliminary assessment of complete blood test when an abnormal number of undifferentiated, premature cells, also known as blasts (lymphoblasts) are found in blood [32] in peripheral blood smear(PBS). To confirm the disease, oncologists recommend bone marrow biopsy by which count and type of blasts cells are identified. More than 20% blasts signifies the presence of leukemia. The speed of progression is so fast in its acute cases that the patient gets very less time to recovery and succumb to the disease. Thus, the early detection of the disease is of utmost importance. Manual quantification of blast cells is a tedious task and result may vary from one lab to

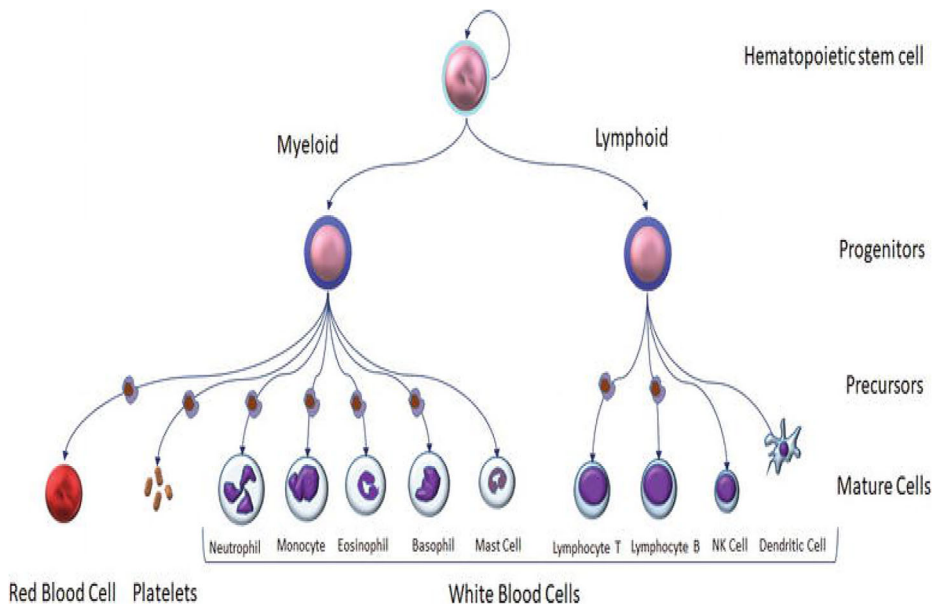


Fig. 1 Types of blood cell components [38]

another and from also from one pathologist to other pathologist. Thus automated screening system, capable of counting the number of blasts to a reasonable accuracy will be an aid to hematologists.

As the normal flow of process in physical medical system is based on collecting blood samples and scanning of these samples under microscope for quantitative and qualitative assessment of various blood cell components, images of these blood cells may be used for developing computer aided automated Leukemia detection system. Image processing techniques have played very important role in simulating such system. In the pre-Deep learning era, combination of conventional image processing techniques and machine learning(ML) algorithms have played a vital role in such simulation of disease diagnostic models. The steps involved in image processing encompasses a wide range of techniques for image acquisition, enhancement, segmentation, morphological processing, and registering etc. [11] ML algorithms are used after extracting features from segmented images for faster and effective classification. As the type and set of features extracted may vary from researcher to researcher and is also dependent on the quality of segmentation of images, the classification accuracy using primitive machine learning algorithms could not reach level of satisfaction [49].

Deep learning has evolved as a new dimension of computer vision [48]. The traditional set of segmentation and conventional Machine Learning algorithms has now been replaced by deep learning techniques as these techniques are robust and capable of performing the dual task of feature extraction from the images and classification of input images into various classes, automatically and efficiently. Deep learning techniques have now become the new normal for image processing and classification task.

Motivation There are many deep neural networks(DNNs) available which are already pre-trained on very large datasets during ImageNet Large-Scale Visual Recognition Challenges

Table 1 Comparison of free parameters in DNNs

Model Name	No. of free learnable parameters
AlexNet	62,378,344
VGGNet	138,423,208
ResNet 18	11,511,784
Inception Net	6,414,360
ResNet 152	60.3 millions
Proposed Net	41,626

(ILSVRC), which have has been contested every year since 2010 and are capable of recognizing upto 1000 categories. In addition to this, they can be re-trained on a different dataset using transfer learning for classifying images in a new domain into desired classes, but those networks are so complex, computation expensive and requires days to be trained for new dataset. Besides this, the number of free learnable parameters to be tuned by the existing DNNs like AlexNet, VGGNet, ResNet-50, ResNet-152, InceptionNet etc is in millions, while our proposed architecture requires to tune only 41,626 free parameters for training on current dataset. The detailed comparison of free parameters has been described in Table 1.

To counter these research gaps, we propose in this paper, a novel deep learning framework (DLF) for the efficient classification of blood cell images through images processing into malignant lymphoblasts and benign lymphocytes. The proposed system is capable of efficiently classifying the blood cell images with an excellent accuracy, even with less training time and less number of learnable free parameters, into healthy and diseased classes.

The remainder of this paper is divided into 5 sections. Section 2 summarises related studies and work in the field. Section 3 describes the deep learning approach and the framework used. Section 4 gives an insight into the implementation. Section 5 provides the experimental results and discussion while Section 4 concludes the paper and future scope of work in the same field.

2 Related work

2.1 Machine learning algorithms

ML algorithms have the exceptional capabilities to learn from the data. These algorithms can learn the patterns or features hidden in the underlying dataset and hence are capable of classifying the data into various categories successfully. These algorithms can also be used for regression problems and statistical analysis. ML algorithms have found a wide spread of application domains e.g. Uber, GoogleMaps, smartphone applications, SnapChat, Computer Vision [45], robust face detection applications [52, 53], pattern classification [6, 7], and classification of ECG signals [16]. Medical diagnosis is no exception. ML algorithms have been successfully used for developing predictive models for cancer detection as well [15, 23], breast Cancer detection [17, 26], and study of multiple myeloma [55] etc. Various ML algorithms have been used in the literature for the leukemia detection [23, 30, 54]. The examples include K-Nearest Neighbour(K-NN) [34, 37], Multi-layer Perceptron(MLP) [54], Support Vector Machine(SVM) [1, 35, 36, 46], Artificial Neural Network(ANN) [50], Bayesian Networks(BN) [2], Ensemble Networks [54] and Random Forests etc. [29]

After the Convolution Neural Networks(CNN) came into existence, we can observe a descend in the usage of machine learning algorithms for image classification. There are two prominent reasons for it:

1. *No need of feature extraction and selection:*

CNN are capable of extracting general and fine features of the images by itself without the implementation of other feature extraction and selection algorithms

2. *Performance:* CNNs have demonstrated higher accuracy on image data, compared to ML algorithms

2.2 Convolution neural network

Deep learning is generally implemented through Convolution Neural Networks(CNN) which are specialized neural networks with many hidden layers, along with the input and the output layers. The term deep represents the deepness of hidden layers. Unlike neural networks, that employ general matrix multiplication for weight calculation or learning, CNN involve convolution operation, which is a special linear mathematical operation, hence the term convolution neural network is used. A CNN is a regularized form of multi layer perceptrons(MLP). A CNN exploits the hierarchical pattern existing in the input data and each of its convolution layer is used to extract these local patterns to build up more complex pattern or feature-map from earlier obtained local patterns. The deep learning has proven to be a boon in the field of image classification as it has been in use for object detection for various applications. There are various pre-defined networks which have been designed and trained extensively for the purpose of object detection and other computer vision tasks e.g., AlexNet, GoogLeNet, ImageNet, ResNet etc. and later modified by researchers to suit their problem at hand by the method known as transfer learning [5, 19, 33]

The two commonly used approaches with the Deep Learning are:

1. Creating Deep learning networks from scratch and training them rigorously on the images to learn the pattern by itself so that the net is able to find the pattern in the unseen images and performs the classification [21, 22]
2. Transfer learning: in which the pre-trained networks are picked up and fine tuning of hidden layers is performed to suit the individual requirement [31, 54].

Deep learning networks have been used over a broad range of applications. e.g, thyroid nodules detection [56], gene expression analysis, biomedical image processing, biomedical signal processing etc. Shafique et al. [42] implemented transfer learning on pre-trained deep neural network, AlexNet for the classification of blood cell images into healthy and diseased, and also for the identification of sub-types of ALL. They achieved 99.5% accuracy for ALL detection and 96.06% accuracy for ALL sub-type classification. In order to achieve this accuracy, they performed image augmentation to reduce over fitting and used other image pre-processing techniques as well. Shahin et al. [44] proposed the deep neural network, WBCNet for the differential identification of WBCs (Lymphocyte, Monocyte, Eosinophil, Basophil and Neutrophil) in blood cell images and achieved 96.1% accuracy on 1268 images. They followed a lengthy procedure of WBC segmentation into nucleus, cytoplasm and cell region followed by feature extraction, feature selection before training on CNN. Thanh et al. [49] proposed a CNN to differentiate the blood cells into normal and abnormal and achieved 96.6% accuracy after reducing the input volume size and applying various types of augmentations to input images. Sipes et al. [47] proposed a CNN for image

based classification of ALL. Before providing the images as input to the proposed CNN, they pre-processed the images using mean cell image subtraction technique to improve model training and accuracy. The author achieved 88% accuracy. Ahmed et al. [4] designed a CNN architecture to differentiate the cells into healthy and diseased and also subtype classification. They used 5-fold cross validation and could achieve 88.25% accuracy in the first category. M.Claro et al. [13] performed the leukemia classification on a collection of 16 dataset containing 2415 images using CNN(Alert Net, Alert Net-R, AlertNet-X, inspired by existing pre-trained deep networks e.g.AlexNet, ResNet, Xception Net respectively and achieved maximum accuracy of 97.18% accuracy. Md. Safuan et al. [39] performed ALL detection on ALL-IDB2 dataset using AlexNet, GoogleNet, and VGG-16 and obtained the accuracy for the identification of lymphoblasts and non-lymphoblasts as 97.74% and 95.29% respectively. Maneela Shaheen et al. [43] proposed a CNN, based on AlexNet for AML detection from 4000 images and obtained 98.58% accuracy.

After performing a detailed study of related work in this domain, we find that there is still a gap in the designing of a simple, but efficient deep network architectures which can provide a comparable accuracy as that of existing deep neural networks. Though the existing DNNs can efficiently perform classification and other computer vision related tasks with high accuracy, but are computationally quite exhaustive as well as expensive and take days to tune millions of free parameters while training. In this work, the authors propose a novel, deep layer based framework that is meticulously trained on leukemia dataset to give exceptional results. Various experiments were performed with different number and types of layers before finalising the final model. The network, so obtained is simple and need approximately, 1400 times less free parameters to be tuned and hence is extremely fast as compared to other existing DNNs. The next section gives an insight into the implementation aspects of the proposed architecture.

3 Proposed work

3.1 Data set used

Since the medical data holds privacy issue, so obtaining publicly available data set for biomedical applications is a challenging task. Fabio Scotti et al. [40] have captured images and created a data set, named ALL-IDB. They made it publicly available for research community for experimentation purposes. The data set consists of two sub-sets. The first one, named ALL-IDB1 contains 108 peripheral blood cell microscopic images captured with optical laboratory microscope with a Canon Powershot G5 camera. The images, have the color depth 24-bit, and a resolution of $1368 \times 1712 \times 3$. These images comprises of 39,000 blood cell elements, including 510 candidate lymphocytes. Out of these 108 images, 59 belongs to healthy human beings and 49 belong to leukemic patients [14, 24, 25]. The second data set, named as ALL-IDB2, consists of 260 segmented images consisting of 130 healthy and 130 diseased lymphoblasts. These images have been cropped from ALL-IDB1. All the images are labelled by expert hematologists.

3.2 Proposed framework

Inspired by the characteristic features of deep learning methods, we propose a novel deep learning framework(DLF) based on CNN to differentiate the normal leukocytes from malignant lymphoblasts. The CNN, like any other neural network, uses two mechanisms:

1. Forward propagation- in which input is taken into the network, processed by applying some mathematical functions to finally generate the output.
2. Backward propagation- in which the output generated at the output layers is compared with the desired output and the difference is calculated as error which is then propagated backwards to adjust the parameters that affects the final output.

CNN utilizes three types of layers, convolution layers, pooling layers and fully connected layers. The input passes through the first convolution layer which extract the local spatial features map. The feature map, thus obtained is very large. To reduce its size, pooling is used. The same process is applied multiple times to fetch the finest local features based on the filters used in each convolution layer and finally the feature-map is given as input to fully connected layer that actually performs classification.

In this work, we designed a CNN containing 16 layers, as represented by framework shown in Fig. 2 below.

In this framework, first layer is the input layer. Next 12 layers perform feature extraction and the other 3 layers (fully connected, softmax and Classification) classify image on the basis of the extracted features. The description of these layers is summarised below in Table 2.

The feature extraction layers consists of three sets of the three layers i.e., convolution layer, batch normalization layer, Relu layer with MaxPool layer following each set. We have represented this set of three layers by a block as given in Fig. 3.

The complete architecture is presented in Fig. 4. The details of various layers in the framework is mentioned below:

3.2.1 Convolution layer

The input image has the size of 200x200x3. In general, for a layer represented by the input dimensions

$$D_n * D_n$$

and a filter having dimension

$$D_f * D_f,$$

, the stride, given by s and padding, given by P , the size of the output is given by:

$$\left[\frac{(D_n - D_f + 2P)}{s} + 1 \right], \left[\frac{(D_n - D_f + 2P)}{s} + 1 \right] \quad (1)$$

The first convolution layer has a filter size of 3x3 and uses a total of 8 different filters. The stride is 1 and ‘same’ padding is applied. The second and third convolution layers have the same structure as the first one but uses different number of filters, 16 and 32, respectively. We use pooling layer with filter size 2, stride 2 and zero padding to decrease the volume spatially. During the learning, the size of the mini batch is kept at the default value i.e. 256.

3.2.2 ReLu layer

The rectified linear activation function i.e., ReLu [20] is a piece-wise linear function which gives output equal to the input, provided the input is positive, else provides zero as output. Since it doesn’t suffer from gradient loss problem like sigmoid and hyperbolic tangent activation functions, it has become the default standard for the activation function for deep



Fig. 2 Deep Learning Framework

networks. We have also used ReLu as the activation function in each feature extraction block. The equation for the ReLU activation function can be described as:

$$ReLU(x) = \max(0, x) \quad (2)$$

3.2.3 Batch normalization layer

Batch normalization is a technique that stabilizes the “internal covariate shift” i.e input distribution to deep layers. in other words, it standardizes the inputs to the next layer in the network to one of the ranges $[0,1]$, $[-1,0]$ or mean $(\mu_B) = 0$ and variance $(\sigma_B) = 1$. It also accelerate the training, in some cases by halving the epochs or better, and reduces the need

Table 2 Description of various layers in the proposed framework

Layer	Type	Activations	Learnables
Input $200 \times 200 \times 3$ images with 'zerocenter' normalization	Image Input	$200 \times 200 \times 3$	–
Conv-1 $8 (3 \times 3 \times 3)$ convolutions with stride[1,1] and padding 'same'	Convolution	$200 \times 200 \times 8$	Weights $3 \times 3 \times 3 \times 8$ Bias $1 \times 1 \times 8$
BN-1 Batch normalization with 8 channels	Batch Normalization	$200 \times 200 \times 8$	Offset $1 \times 1 \times 8$ Scale $1 \times 1 \times 8$
ReLU-1 ReLU	ReLU	$200 \times 200 \times 8$	–
MaxPool-1 2×2 max pooling with stride [2,2] and padding [0 0 0 0]	Max Pooling	$100 \times 100 \times 8$	–
Conv-2 $16 (3 \times 3 \times 8)$ convolutions with stride[1,1] and padding 'same'	Convolution	$100 \times 100 \times 16$	Weights $3 \times 3 \times 8 \times 16$ Bias $1 \times 1 \times 16$
BN-2 Batch normalization with 16 channels	Batch Normalization	$100 \times 100 \times 16$	Offset $1 \times 1 \times 16$ scale $1 \times 1 \times 16$
ReLU-2 ReLU	ReLU	$100 \times 100 \times 16$	–
MaxPool-2 2×2 max pooling with stride [2,2] and padding [0 0 0 0]	Max Pooling	$50 \times 50 \times 16$	–
Conv-3 $32 (3 \times 3 \times 16)$ convolutions with stride[1,1] and padding 'same'	Convolution	$50 \times 50 \times 32$	Weights $3 \times 3 \times 3 \times 16 \times 32$ Bias $1 \times 1 \times 32$
BN-3 Batch normalization with 32 channels	Batch Normalization	$50 \times 50 \times 32$	Offset $1 \times 1 \times 32$ scale $1 \times 1 \times 32$
ReLU-3 ReLU	ReLU	$50 \times 50 \times 32$	–
MaxPool-3 2×2 max pooling with stride [2,2] and padding [0 0 0 0]	Max Pooling	$25 \times 25 \times 32$	–
FC-1 2 fully connected layer	Fully Connected	$1 \times 1 \times 2$	Weights 2×20000 Bias 2×1
SoftMax softmax	Softmax	$1 \times 1 \times 2$	–
Classification cross-entropy with classes 'blast' and 'normal'	Classification Output	–	–

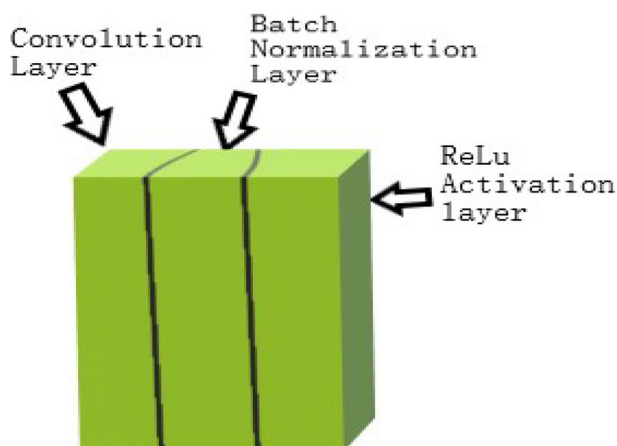


Fig. 3 Block Architecture

of regularization e.g. L2 regularization or dropouts, reducing generalization error [27]. Thus we have performed batch normalization of the input data before forwarding it to activation layer. The equation for mean and variance for the layer input may be given by:

$$\mu_B = \frac{1}{m} \sum_{k=1}^m x_k \quad (3a)$$

$$\sigma_B^2 = \frac{1}{m} \sum_{k=1}^m (x_k - \mu_B)^2 \quad (3b)$$

Next, the layer inputs are normalized using earlier calculated statistics, represented as $\overline{x_k}$ given by:

$$\overline{x_k} = \frac{x_k - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \quad (3c)$$

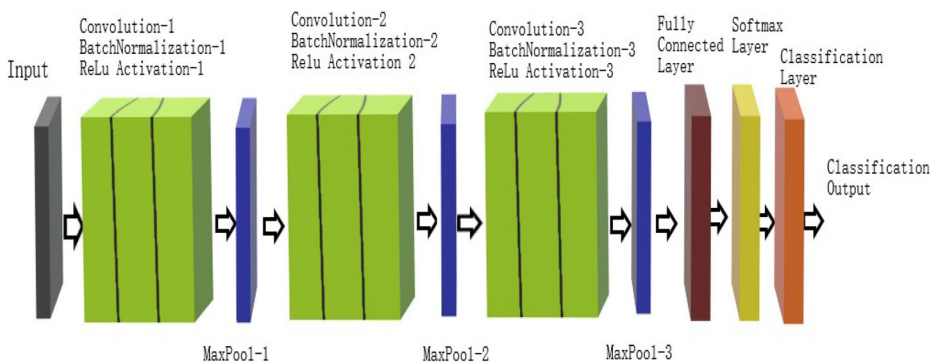


Fig. 4 CNN Architecture

Finally scale and shift is performed to obtain the final output of the layer

$$y_k = \gamma \cdot \overline{x_k} + \beta \quad (3d)$$

where, y_k is the final pre-processed input to the next layer. γ , the scale parameter and β , the shift parameter, are among the other parameters of the network that are learned during training.

3.2.4 Max-pool layer

Convolution layer having m kernels may learn to detect up to m local features. The feature map generated after each set of convolution layers is huge in size and complexity. This complexity may be reduced by using a pooling layer after each set of convolution layers [48]. The pooling layer reduces the number of parameters and the complex computations in the networks, thus reducing the size of feature map spatially. The most commonly used pooling method is Max-pooling, followed by Average-pooling and Global-pooling. The pooling operation is performed by sliding a 2-Dimensional filter over each channel of feature map and summarising the features lying under the region covered by the filter [44]. Given a feature map with the dimensions $n_h \times n_w \times n_c$, the dimensions of output obtained after a pooling layer is given by equation:

$$\frac{n_h - f + 1}{s_f} \times \frac{n_w - f + 1}{s_f} \times n_c \quad (4)$$

where,

- n_h - height of feature map
- n_w - width of feature map
- n_c - number of channels in the feature map
- f - size of filter
- s_f - stride of filter

To achieve the computational efficiency in modern CNNs, the pooling layer is replaced by a convolution layer with a stride greater than 1.

3.2.5 Fully connected layer

The features output by the convolution layer, after being down-sampled by the pooling layer, are fed into one or more fully connected layers. As implied by the name, all neurons in a fully connected layer are connected to all the neurons in the preceding layer. This layer combines all of the local patterns learned by the previous feature selection layers across the image to recognize the larger patterns in order to perform classification. For this reason, the output-size of the last fully connected layer of the network is equal to the desired number of classes of the data set. The fully connected layer accepts only 1-Dimensional(D) input, so the output of the previous layer is first converted to a 1-D vector. This layer performs matrix multiplication between the input and a weight matrix and then adds a bias vector to the result, which is also known as *linear transformation* of the input data and is given by (5) as:

$$Z = W^T \cdot X + b \quad (5)$$

3.2.6 Softmax layer

The Softmax layer is the penultimate layer of the CNN. It applies a softmax function to the vector input generated by fully connected layer and converts the scores to normalized probability distribution. The softmax function is also known as multi-class logistic regression because it is a generalized logistic regression that may be applied to multi-class classification problems. The softmax function can be used for the classification, only when the classes are mutually exclusive. The equation of the softmax function is:

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (6)$$

where,

$\mathbf{z}$ = The vector input to the softmax function, made up of $(z_0, \dots, z_K)$,

z_i = All the z_i values i.e., $(z_0, \dots, z_K)$ are the elements of the input vector and may take up any real value and,

k = Number of terms.

3.2.7 Classification layer

The last layer following the Fully connected layer and the Softmax layer is the final output layer and hence is called as the classification layer. Number of neurons in the final output layer is equal to the number of categories for classification problem at hand.

3.3 Experiment and results

Our experiments were conducted on MATLAB 2020b on an Intel core i5 processor with 2.53 GHz processing speed. After experimenting with various architecture models of CNN, we zeroed in on a deeper 16 layer CNN architecture which resulted in 100% accuracy in maximum no of repetitions during the experiment. The experiment was conducted on ALL-IDB1 data-set, which is described in in Section 3.1. The dataset used consists of 108 images among which 59 belong to normal healthy category and 49 belong to abnormal diseased category. The input image was resized to $(200 \times 200 \times 3)$ from $(1368 \times 1712 \times 3)$ to make the computation faster and memory efficient. The resized image was then given as input to the 16 layer deep convolution neural network. The dataset was divided into training and testing parts and the experiment was conducted with different variations of the training and testing images by percent and by number of images i.e. (80%,20%), (60,48) etc. The experiment was conducted with different number of epochs to check the speed and robustness of the model. We did not need to conduct any special feature extraction in this case as we did in our previous work [10] as CNN is capable of extracting all the necessary features on its own. The Fig. 5 demonstrate the training of CNN for one execution of the data-division as (80%,20%) using 20 epochs, clearly indicating the validation accuracy as 100% and training time as 1 minute 43 seconds.

The final accuracy of the model is given after averaging various accuracy of the model for various division of data and various number of epochs. The proposed CNN resulted in an average accuracy of 97.954% for 20 executions of the experiment with the division of data into 80% training and 20% testing data for 20 epochs(Experiment-A). When the dataset was divided by number of images, i.e., 60 training and 48 testing images for 30 epochs

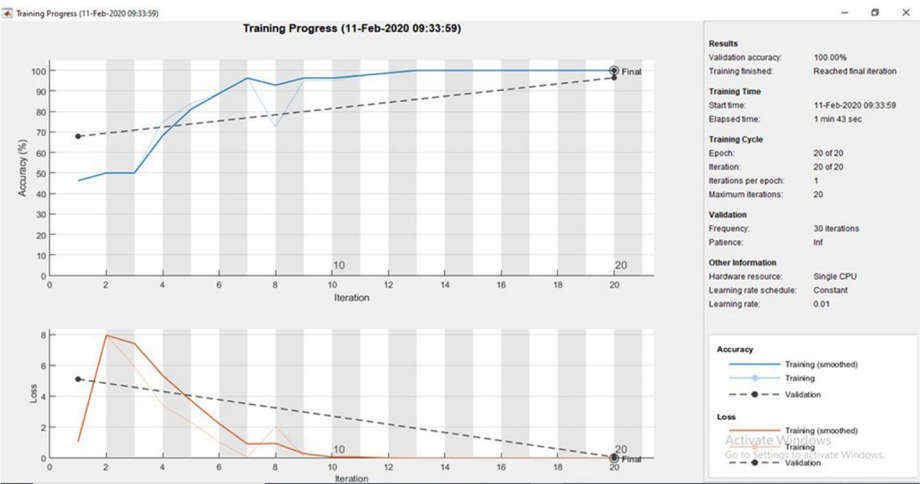


Fig. 5 Demonstration of one execution of training of CNN Model for 80%, 20% data division and 20 epochs

(Experiment-B), the average of 6 randomly selected execution gave an accuracy of 98.62% as shown in Table 4. To find the overall average we took into account, 6 random executions Experiment-A as shown in Table 3 with average accuracy 97.73%, and the average obtained from Experiment-B and finally the super average of above two average values was calculated, which came out to be 98.17%. The detailed experiment specifications are presented in Tables 3 and 4.

Performance evaluation The performance of the results thus obtained, are evaluated on the basis of accuracy and confusion matrix. The confusion matrix depicts the values of the metrics True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) as demonstrated in the general structure of confusion matrix in Fig. 6, where,

- True Positive (TP): Number of events that are positive and are correctly predicted as positive by the model.

Table 3 Results of Experiment-A [Data division:(80%,20%),epochs=20s]

Execution	No. of mismatch	Accuracy
1	2	90.91
2	0	100
3	1	95.45
4	0	100
5	0	100
6	0	100
Average Accuracy		97.73%

Table 4 Results of Experiment-B [Data division:(60,48), epochs=30s]

Execution	No. of mismatch	Accuracy%
1	0	100
2	0	100
3	1	97.92
4	3	93.79
5	0	100
6	0	100
Average Accuracy		98.62%

- False Positive (FP): Number of events that are negative but are incorrectly predicted as positive by the model.
- True Negative (TN): Number of events that are negative but are correctly predicted as negative by the model.
- False Negative (FN): Number of events that are not negative but are incorrectly predicted as negative by the model.

The Accuracy is described in terms of TP, FP, TN, FN as shown in the (7) as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

The values of TP, TN, FP and FN obtained for a single execution of experiment is demonstrated in Fig. 7.

A comparison of the work proposed by the contemporary researchers has been drawn into Table 5.

Confusion Matrix		
	blasts	normal
blasts	TP	FN
normal	FP	TN

Fig. 6 General Confusion matrix showing the TP, TN, FP and FN

Confusion Matrix		
	blasts	normal
blasts	18	1
normal	0	29

Fig. 7 Confusion matrix for one experimental result

Limitations of the proposed study The proposed study is focused on a small, publicly available balanced dataset. The efficiency of the proposed framework has not been tested over other datasets.

4 Conclusion

This paper has demonstrated and established that convolution neural network has given the best results for the image data, as compared to conventional machine learning algorithms as evident in the literature. A novel deep learning framework is proposed for detecting acute lymphoblastic leukemia by classifying the microscopic blood cell images into healthy and diseased. The salient features of the framework are:

1. There is no need of feature extraction and selection as demanded by conventional ML algorithms.
2. The number of free parameters required to be tuned for the training of proposed framework model is approximately 1400 times less as compared to the existing pre-trained deep models, which came into existence after participation in ImageNet Large-Scale Visual Recognition Challenge(ILSVRC) conducted in recent years and is capable of achieving superior accuracy.
3. The proposed DLF is quite simple in structure and takes very less time for training. The reason for this is small sized kernels used in convolution layers and the batch normalization that reduces the requirement of regularization. Our proposed DLF takes few minutes for training whereas other state-of-the-art networks takes days to be trained, which is an exemplary achievement.

The classification of 108 images of leukemia using the proposed DLF has provided efficient results with 100% best accuracy. The experiment is conducted rigorously on a simple i5, 1.53GHz processor and 4GB RAM. The experiments have been conducted over many repetition of training and testing data, with different bifurcations, with training conducted for different number of epochs. The average accuracy of two experiments i.e. Experiment-A and B is obtained by the average of any 6 random repetitions in those experiments and

Table 5 Comparison of proposed work with other state of the art work

Ref.	Work	Year	No. of Images	Features	Classifier	Accuracy
[44]	Shahin et al.	2019	2551	Deep Features	WBCNet	96.1%
[51]	Thanh et al.	2017	1188(on Augmentation)	Deep Features	CNN	96.6%
[37]	Rehman et al.	2018	330	Deep Feature + LBP Features	CNN	97.78%
[47]	Sipes et al.	2018	388	Deep Features	CNN	88%
[41]	Shafique et al.	2018	760(on Augmentation)	Deep Features	CNN	99.5%
[4]	Ahmed et al.	2019	2478(on Augmentation)	Deep Features	CNN	88.25%
[13]	M.Claro et al.	2020	2415	Deep Features	CNN	97.18%
[3]	R. Agarwal et al.	2019	100	GLCM	CNN	97.3%
[39]	Mohd. Safuan et al.	2020	260	Deep Features	AlexNet	97.74%
[43]	Maneela Shaheen et al.	2021	4000	Deep Feature	AlexNet LeNet	98.58% 96.25%
Proposed Method	Sunita Chand et al.	2021	108	Deep Feature	Proposed DLF	100%

is found to be 97.73% (for the training:testing data= 80%:20%, 20 epochs) and 98.62% (for the training:testing data = 60 images:48 images, 30 epochs).

Future scope Though the proposed DLF has been trained on a smaller balanced dataset, the same network may be tested for other dataset(balanced as well as unbalanced) to check its efficiency and effectiveness on the same problem domain as well as other domains in the future work. Also the research work can be extended to detect all sub-types of ALL, all 4 types of leukemia, detection of AML etc. using the same framework or modifications of the proposed framework.

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Declarations

Competing interests The authors declare that they have no conflict of interest.

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